



histopathology Inflammation part I

3rd class Lecture 4

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Objectives

1. Recall the key cardinal signs of inflammation and list at least four causes of acute inflammation.
2. explain the stepwise vascular and cellular responses involved in acute inflammation, including changes in flow, permeability, and leukocyte migration.
3. Identify whether inflammation is acute or chronic based on clinical signs, cellular components, and histologic features.
4. Analyze the phases of leukocyte recruitment and identify specific molecules involved in rolling, adhesion, and transmigration processes.
5. Evaluate the roles of various chemical mediators (e.g., histamine, prostaglandins, cytokines) in modulating vascular reactions and leukocyte behavior in inflammation.
6. Construct a concept map connecting major cell-derived and plasma-derived mediators of acute inflammation, including their sources, targets, and effects.

Inflammation

Inflammation: is a response of vascularized tissues to infections and tissue damage that brings cells and molecules from the circulation to the sites where they are needed, to eliminate the offending agents.

- It is actually a protective response.
- It serves to eliminate:
 - The **initial cause** of cell injury (e.g., **microbes, toxins**).
 - The **consequences** of injury (e.g., **necrotic cells** and tissues).
- The **mediators of defense** include:
 - Phagocytic leukocytes.
 - Antibodies.
 - Complement proteins.

Inflammation

- The typical inflammatory reaction develops through a **series of steps**:
 1. The **offending agent**, which is located in extravascular tissues, is ***recognized by host cells*** and molecules.
 2. **Leukocytes** and **plasma proteins** are **recruited** from the circulation to the site where the offending agent is located.
 3. The **leukocytes** and **proteins** are **activated** to destroy and eliminate the offending substance.
 4. The reaction is **controlled** and **terminated**.
 5. The damaged tissue is **repaired**.

Types of inflammation

- Inflammation may be of two types, **acute** and **chronic**.

I. Acute inflammation: The initial, rapid response to infections and tissue damage.

- It typically develops within **minutes** or **hours**.
- is of **short duration**, lasting for several **hours** or a **few days**.
- Its main characteristics:
 - edema: the **exudation** of **fluid** and **plasma proteins**.
 - the ***emigration of leukocytes***, predominantly **neutrophils**.
- When it achieves its desired goal: the **reaction subsides** and residual injury is **repaired**.
- if the initial response fails to clear the stimulus: *the reaction progresses* to **chronic inflammation**.

II. Chronic inflammation:

- may **follow** acute inflammation or arise **de novo**.
- Characterized by:
 - It is of longer duration.
 - is associated with ***more tissue destruction***.
 - the presence of **lymphocytes** and **macrophages**.
 - the proliferation of **blood vessels**.
 - **fibrosis**.

Table 3.1 Features of Acute and Chronic Inflammation

Feature	Acute	Chronic
Onset	Fast: minutes or hours	Slow: days
Cellular infiltrate	Mainly neutrophils	Monocytes/macrophages and lymphocytes
Tissue injury, fibrosis	Usually mild and self-limited	May be severe and progressive
Local and systemic signs	Prominent	Less

– The **external manifestations** of inflammation (**cardinal signs**) are:

1. Heat (calor in Latin)

2. Redness (rubor)

3. Swelling (tumor)

4. Pain (dolor)

5. Loss of function.

ST the **inflammatory** reaction becomes the cause of disease:

1. inflammatory reactions to infections:
 - The **harmful consequences** are **self-limited** and **resolve** as the inflammation abates.
2. inflammatory reaction is misdirected (**autoimmune diseases**).
3. inflammatory reaction is occurs against substances that evoke an immune response (**allergies**).
4. inflammatory reaction is excessively prolonged (**microbes resist eradication**).
5. Inflammatory reactions underlie common **chronic diseases**:
 - Rheumatoid arthritis, Atherosclerosis, Lung fibrosis.
 - life-threatening hypersensitivity reactions to
 - insect bites, drugs, and toxins.

- Inflammation **is terminated** when the **offending agent** is **eliminated**.
 1. **Mediators** are broken down and dissipated.
 2. **Leukocytes** have short life spans in tissues.
 3. **Anti inflammatory** mechanisms are activated.

- After inflammation has **terminated**, the process of **tissue repair** start.

- **Repair**: a series of events that heal damaged tissue.
- the injured tissue is replaced through:
 1. **Regeneration** of surviving cells.
 2. Filling of residual defects with connective tissue (*scarring*).

CAUSES OF INFLAMMATION

- Inflammatory reactions may be triggered by a variety of stimuli:
 1. **Infections** (bacterial, viral, fungal, parasitic) and microbial toxins are among the most common and medically important causes.
 2. **Tissue necrosis:** elicits inflammation, which may include
 - *Ischemia* (reduced blood flow, the cause of myocardial infarction).
 - *Trauma*.
 - *physical and chemical injury* (e.g., thermal injury, as in burns or frostbite; irradiation; exposure to some environmental chemicals).
 3. **Foreign bodies** (splinters, dirt, sutures).
 4. **Immune reactions** (hypersensitivity).
 - The immune responses may be directed against self antigens, causing autoimmune diseases.
 - inappropriate reactions against environmental substances, as in allergies, or against microbes.

ACUTE INFLAMMATION

- Acute inflammation has **three major components**:
 1. **Dilation of small vessels**: leading to an increase in blood flow.
 2. **Increased vascular permeability**: enabling plasma proteins and leukocytes to leave the circulation.
 3. **Emigration of the leukocytes from the microcirculation**.
 - Accumulation in the focus of injury.
 - Activation to eliminate the offending agent.

Reactions of Blood Vessels in Acute Inflammation

- The vascular reactions of acute inflammation consist of:
 1. Changes in the **flow** of blood.
 2. Change of vascular **permeability**.
 - to maximize the movement of plasma proteins.
 - Facilitate the movement of leukocytes out of the circulation and into the site of infection or injury.
- **Exudation** : is the escape of fluid, proteins, and blood cells from the vascular system into interstitial tissues or body cavities.

An **exudate**:

- is an extravascular fluid that has a high protein concentration and contains cellular debris.
- **Its presence** implies that there is an increase in the permeability of small blood vessels.

— **Transudate:**

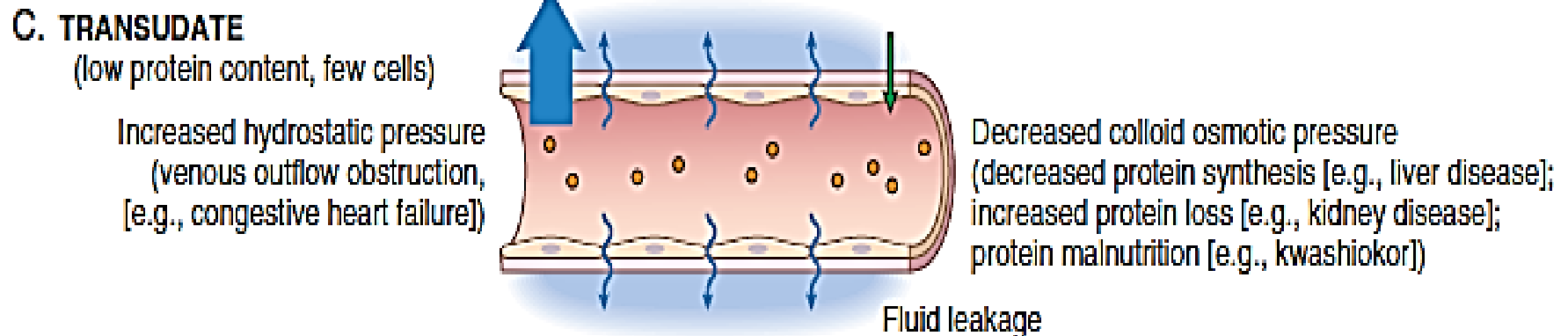
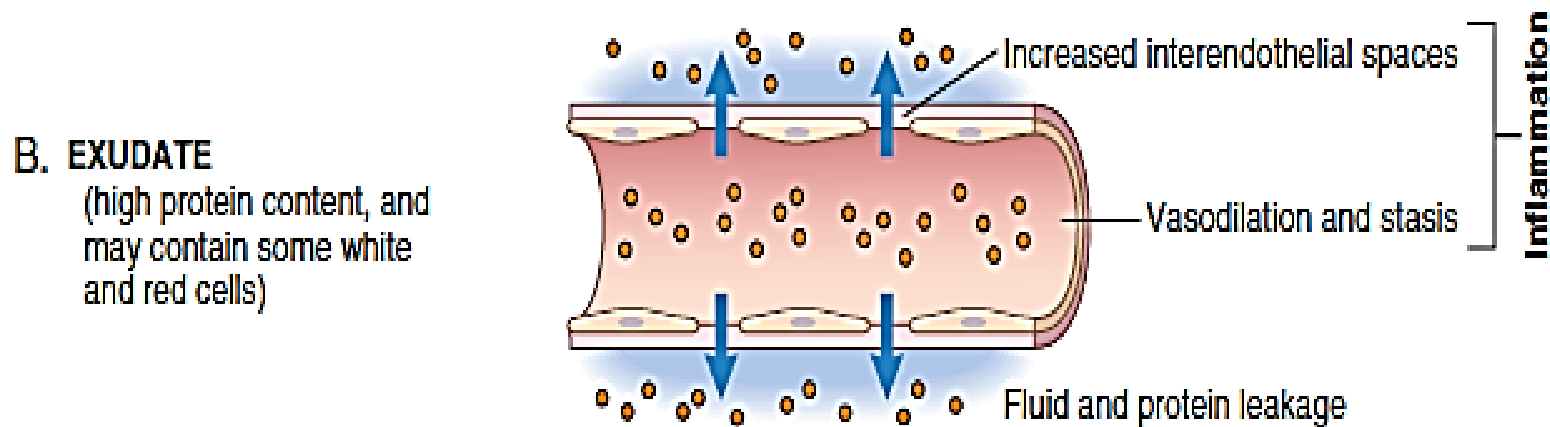
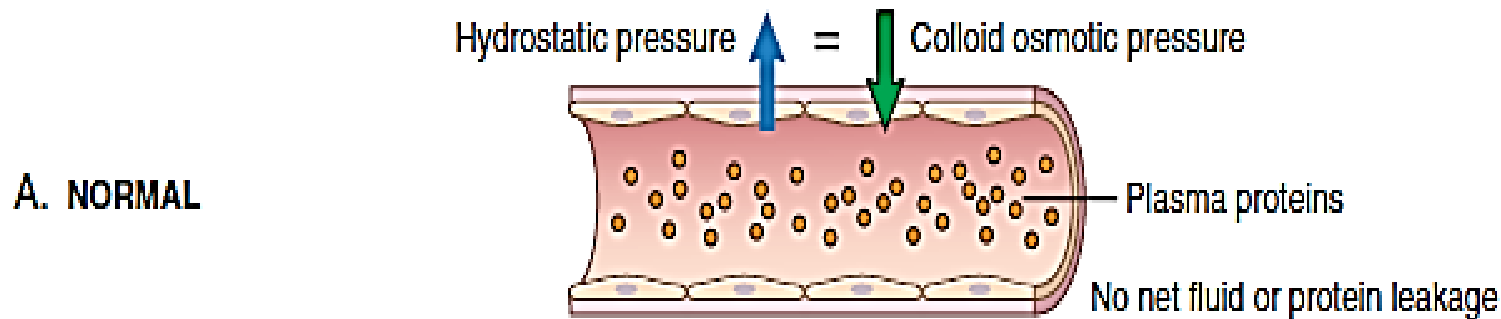
- is a fluid with low protein content,
- little or no cellular material, and
- low specific gravity.
- It is an ultrafiltrate of blood plasma.
- is produced as a result of osmotic or hydrostatic imbalance across vessels.
- **normal vascular permeability.**

— **Edema:**

- an excess of fluid in the interstitial tissue or serous cavities.
- it can be either an exudate or a transudate.

— **Pus:**

- Is a purulent exudate.
- is rich in leukocytes (mostly neutrophils).
- the debris of dead cells, and microbes.



I. Changes in Vascular Flow and Caliber

- Changes in vascular flow and caliber begin early after injury and consist of the following:

1. Vasodilation:

- Is induced by the action of several mediators, **histamine**, on vascular **smooth muscle**.
- It is one of the **earliest manifestations** of acute inflammation.
- May be preceded by **transient vasoconstriction**.
- Vasodilation **first involves the arterioles**.
- Vasodilation leads to the opening of new capillary beds in the area.
- The result is **increased blood flow**, which is the cause of **heat** and **redness** (erythema) at the site of inflammation.

2. Increased vascular permeability:

- with the outpouring of **protein-rich fluid** (an exudate) into the extra-vascular tissues.

3. Stasis of blood:

- The loss of fluid and increased vessel **diameter** lead to:
 - Slower blood flow,
 - Concentration of RBCs in small vessels, and
 - Increased viscosity of the blood.
 - Engorgement of small vessels with slowly moving red cells
- **Histologically:** vascular congestion and
- **Grossly:** localized redness (erythema) of the involved tissue.

- **As stasis** develops
 - blood leukocytes (**neutrophils**) **accumulate** along the vascular **endothelium**.
 - The **endothelial cells** are **activated** by **mediators** produced at sites of infection and tissue damage.
 - The **endothelial cells** express increased levels of **adhesion molecules**.
 - **Leukocytes then adhere to the endothelium**, and soon they **migrate** through the vascular wall into the **interstitial tissue**.

II. Increased Vascular Permeability (Vascular Leakage)

- Several mechanisms are responsible for increased vascular permeability in acute inflammation, which include:

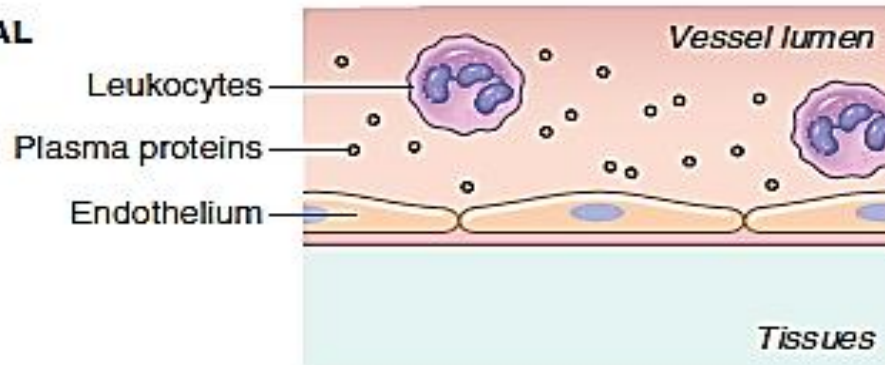
1. Retraction of endothelial cells

- is the most common mechanism of vascular leakage.
- is usually short-lived.
- it is referred to as the immediate transient response.
- Resulting in opening of **inter-endothelial** spaces.
- It is elicited by histamine, bradykinin, leukotrienes, and other chemical mediators.
- It occurs **rapidly** (short lived) after exposure to the mediator (**15-30 minutes**).

2. Endothelial injury:

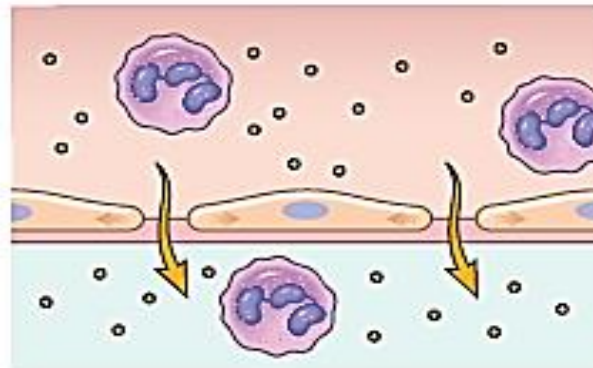
- Resulting in endothelial cell **necrosis** and **detachment**.
- Caused by:
 1. **severe injuries:**
 - Burns
 - the **actions of microbes** and microbial **toxins** on the endothelial cells.
 2. **Neutrophils.**
 - In most instances leakage **starts immediately** after injury.
 - The leakage **sustained** for **several hours (long lived)** until the damaged vessels are thrombosed or repaired.

A NORMAL



B RETRACTION OF ENDOTHELIAL CELLS

- Induced by histamine, other mediators
- Rapid and short-lived (minutes)



C ENDOTHELIAL INJURY

- Caused by burns, some microbial toxins
- Rapid; may be long-lived (hours to days)

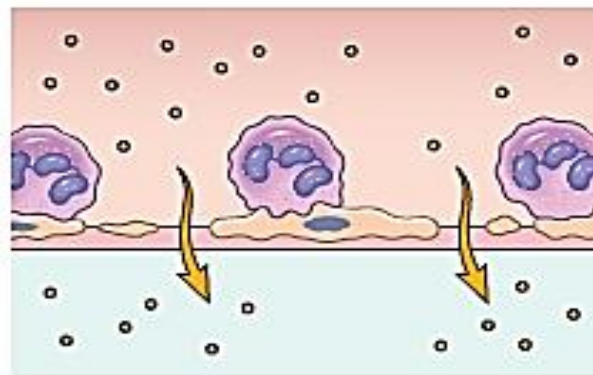


Fig. 3.3 Principal mechanisms of increased vascular permeability in inflammation and their features and underlying causes.

Leukocyte Recruitment to Sites of Inflammation

- **Leukocytes** perform the key function of **eliminating** the offending agents.
- The most important leukocytes in acute inflammatory reactions are:
 - the ones capable of **phagocytosis** (neutrophils and macrophages).
- **Neutrophils:**
 - are produced in the bone marrow
 - rapidly recruited to sites of inflammation.
- **Macrophages:**
 - are slower than neutrophils.
 - more prolonged responders.
 - Macrophages also produce **GF** that aid in **repair**.

- The **journey of leukocytes** from the vessel lumen to the tissue is a **multistep** process that is mediated and **controlled by**:
 1. Adhesion molecules.
 2. Cytokines.
- **Leukocytes** normally flow rapidly in the blood.
- **in inflammation**:
 1. they have to **be stopped** and then
 2. **Brought to the offending agent** or the site of tissue damage, outside the vessels.
- **This process** can be divided into phases:
 1. **Adhesion of leukocytes to endothelium** at the site of inflammation.
 2. **Transmigration** of the leukocytes through the vessel wall.
 3. **Movement toward** the offending agent.

I. Leukocyte Adhesion to Endothelium

- Normally red cells, being smaller, tend to move faster than the larger white cells.
- As a result:
 1. Red cells are confined to the central axial column.
 2. Leukocytes are pushed out toward the wall of the vessel, but the flow prevents the cells from attaching to the endothelium.

As the blood flow slows **early** in inflammation (**stasis**):

- A. Hemodynamic conditions change.
- B. More white cells assume a peripheral position along the endothelial surface.

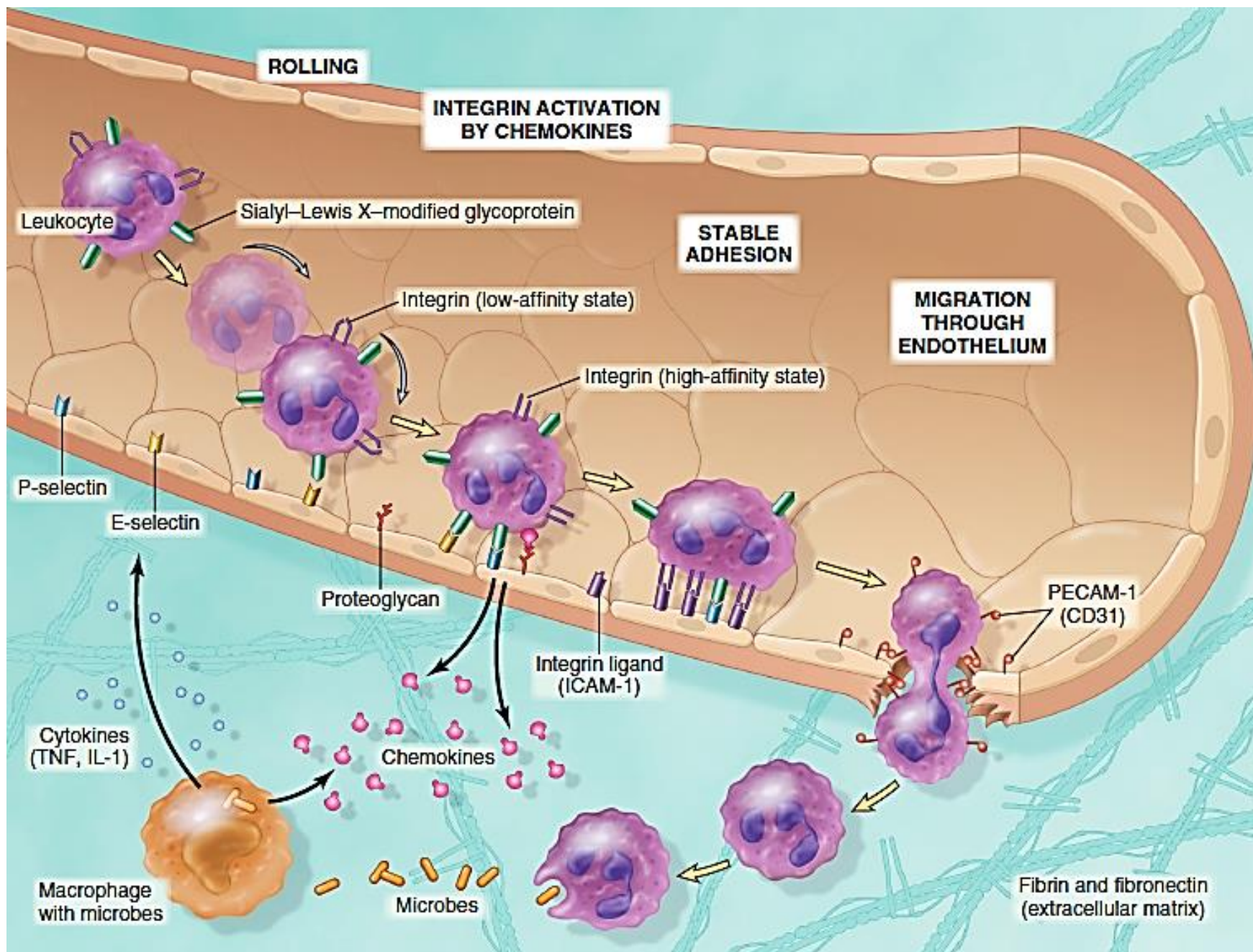
- This process of leukocyte redistribution is called **margination**.

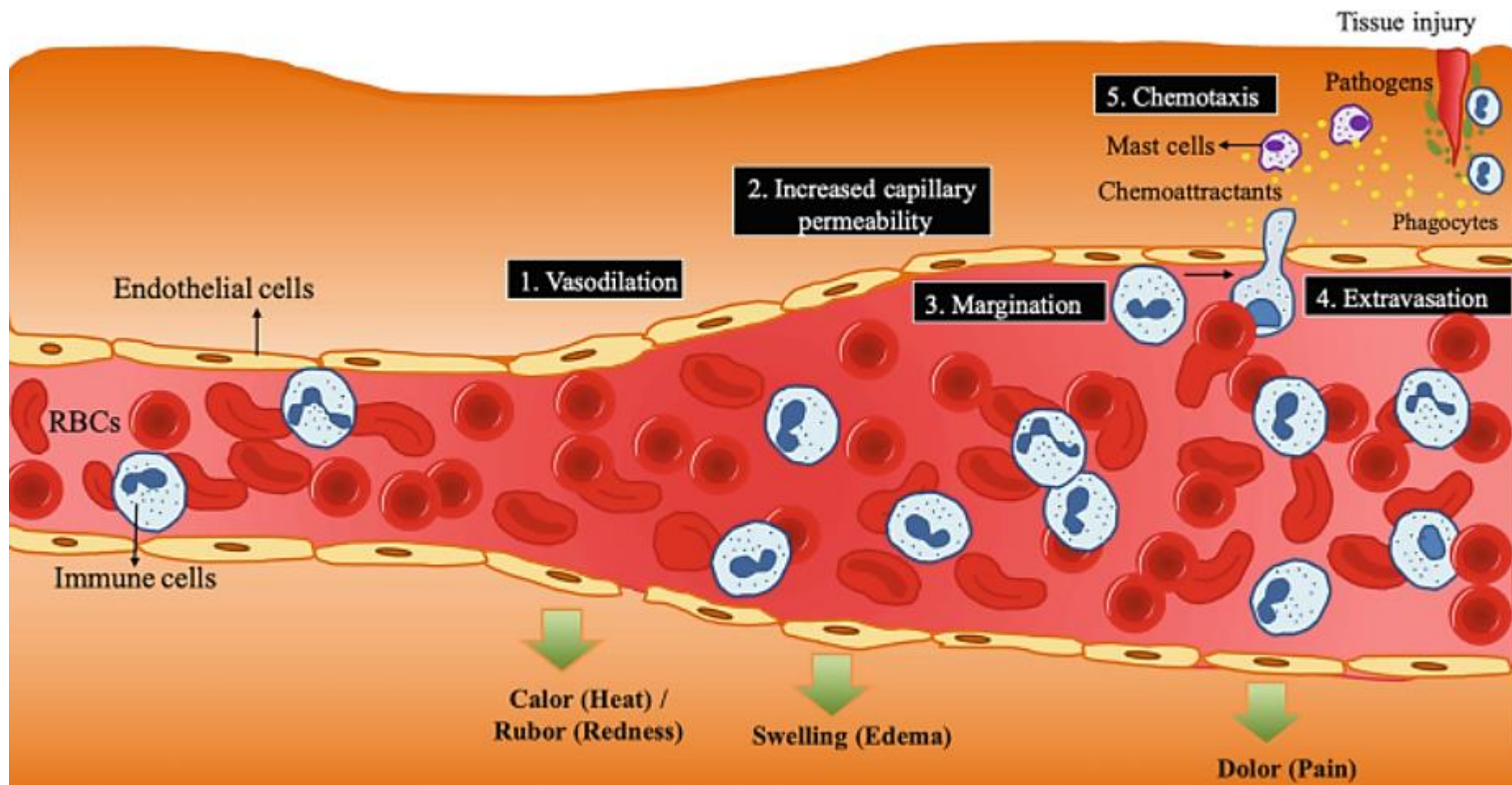
- the **endothelial** cells are:
 - A. **Activated** by **cytokines** and other mediators produced locally.
 - B. Express **adhesion molecules** to which the *leukocytes attach loosely*.
- Leukocytes:
 - bind and detach and thus begin to **tumble** on the endothelial surface, a process called **rolling**.
 - Leukocytes finally come **to rest** at some point where they adhere **firmly**.
- The attachment of leukocytes to endothelial cells is mediated by:
 - **Complementary adhesion molecules** on the two cell types.
 - The expression of adhesion molecules is enhanced by cytokines.

- The **two families of molecules** involved in leukocyte adhesion and migration: (selectins and integrins)
- **Selectins:**
 - Selectins are **receptors** expressed on **leukocytes** and **endothelium**.
 - Mediate the **initial weak** interactions between leukocytes and ECs.
 - The three members of this family are:
 - **E-selectin** expressed on endothelial cells.
 - **P-selectin** present on platelets and endothelium.
 - **L-selectin** on the surface of most leukocytes.
- The endothelial selectins:
 - Expressed at **low levels** or **not at all** on *un-activated endothelium*.
 - **Upregulated** after *stimulation* by cytokines and other mediators .
- Therefore, *binding of leukocytes* is largely **restricted** to the *endothelium* at sites of infection or tissue injury (**where the mediators are produced**).

- **Integrins:**

- Are trans-membrane glycoproteins normally expressed on leukocyte in a low-affinity form.
- Function:
 1. Mediate the **firm adhesion** of leukocytes to endothelium.
 2. Facilitate adhesion of various cells to the extracellular matrix.
- do **not** adhere to their specific **ligands** (on endothelium) until the **leukocytes** are activated by **chemokines** (cytokines that are secreted by many cells at sites of inflammation).
- The **rolling leukocytes** become **activated** after encounter the **chemokines**.
- At the same time, other cytokines, **TNF** and **IL-1**, activate endothelial cells to increase their expression of ligands for integrins.





II. Leukocyte Migration Through Endothelium

- The leukocytes **migrate** through the vessel wall primarily by **squeezing** between cells at intercellular junctions.
- This **extravasation** of leukocytes, called **transmigration**, occurs mainly in post-capillary venules.
- After traversing the endothelium:
 1. leukocytes **pierce** the **basement membrane**.
 2. enter the **extravascular tissue**.
- **Further movement** of leukocytes is driven by **chemokines** produced in extravascular tissues, which stimulate leukocytes to travel along a **chemical gradient**.

III. Chemotaxis of Leukocytes

- Leukocytes move in the tissues toward the site of injury by a process called ***chemotaxis***.
- **Chemotaxis**: is a locomotion along a chemical gradient.
- Both **exogenous** and **endogenous** substances can act as **chemo-attractants**, including the following:
 1. Bacterial products.
 2. Cytokines, especially those of the **chemokine family**
 3. Components of the complement system, particularly **C5a**
 4. Products of (mediators), the lipoxygenase pathway of arachidonic acid (AA) metabolism, particularly leukotriene B4 (**LTB4**).
- The **net result** is that
 1. **leukocytes migrate** toward the inflammatory stimulus.
 2. in the **direction** of the locally produced **chemo-attractants**.

– In most forms of acute inflammation:

1. Neutrophils:

- **Predominate** in the inflammatory infiltrate during the first **6 - 24 hours**.
- Gradually replaced by monocyte-derived macrophages over **24 - 48 hours**.
- After entering tissues:
 - **Neutrophils** are short-lived.
 - Undergo **apoptosis** and **disappear** within **24 - 48 hours**.

2. Macrophages:

- survive longer.
- may **proliferate in the tissues**, and thus they become the **dominant** population in prolonged inflammatory reactions.

- There are **exceptions** to this pattern of cellular infiltration.
 1. Infections produced by ***Pseudomonas* bacteria**—the cellular infiltrate is dominated by neutrophils for several days.
 2. **Viral infections**, lymphocytes are first cells to arrive.
 3. **Hypersensitivity** reactions are dominated by activated lymphocytes, macrophages, and plasma cells (reflecting the immune response).
 4. **Allergic reactions**, eosinophils may be a prominent cell type.

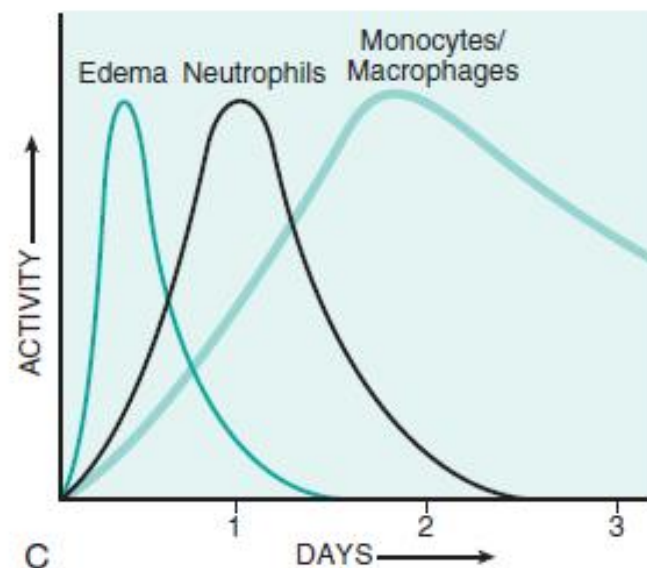
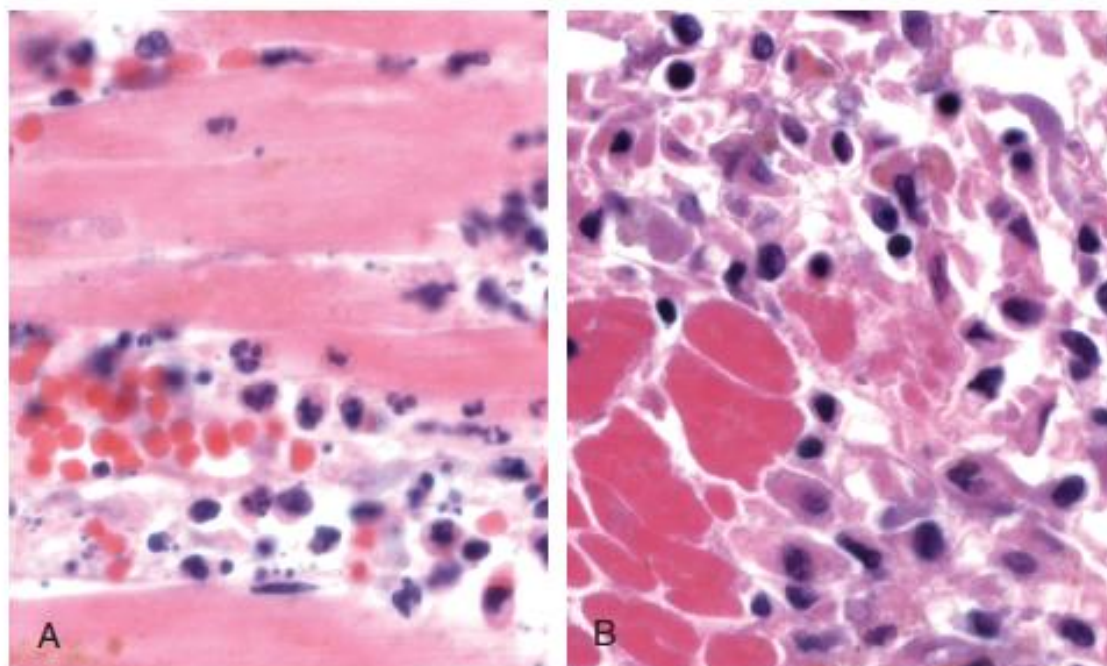


Fig. 3.5 Nature of leukocyte infiltrates in inflammatory reactions. The photomicrographs show an inflammatory reaction in the myocardium after ischemic necrosis (infarction). (A) Early (neutrophilic) infiltrates and congested blood vessels. (B) Later (mononuclear) cellular infiltrates. (C) The approximate kinetics of edema and cellular infiltration. For simplicity, edema is shown as an acute transient response, although secondary waves of delayed edema and neutrophil infiltration also can occur.

IV. Phagocytosis and Clearance of the Offending Agent

- **Phagocytosis** and **intracellular killing**: are the **functional responses** that are most important for **destruction** of **microbes** and other offenders.

Phagocytosis (IN BOOK)

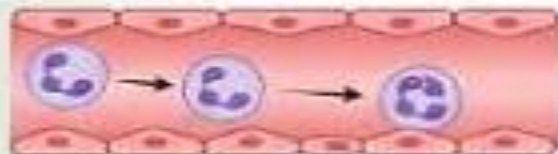
- **Phagocytosis** involves **three sequential steps**:
 1. **Recognition** and attachment of the particle to be ingested by the leukocyte.
 2. **Engulfment** , with subsequent formation of a **phagocytic vacuole**.
 3. **Killing** or degradation of the ingested material.
- These steps are triggered by activation of phagocytes by microbes, necrotic debris, and various mediators.

LEUKOCYTE RECRUITMENT:

From Margination to Phagocytosis

1 MARGINATION

- White blood cells move toward the edge of blood flow.



2 ROLLING

- Neutrophils loosely bind and roll along the vessel wall.



3 ADHESION

- Firm attachment to endothelial cells.



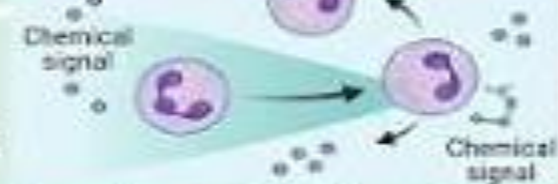
4 EMIGRATION

- Squeezing through the vessel wall (Diapedesis).



5 CHEMOTAXIS

- Migration toward the injury/infection site along a chemical gradient.



6 PHAGOCYTOSIS

- Engulfing and destroying bacteria or cellular debris.



Margination

Rolling

Adhesion

Emigration

Chemotaxis

Phagocytosis

MEDIATORS OF INFLAMMATION

- **The mediators of inflammation:** are the **substances** that **initiate** and **regulate** inflammatory reactions.
- The most important mediators of acute inflammation are:
 1. Vasoactive amines.
 2. Lipid products (prostaglandins and leukotrienes).
 3. Cytokines (chemokines).
 4. Products of complement activation.
- Mediators may be produced:
 - A. Locally:** by cells at the site of inflammation.
 - B. Circulating inactive precursors:** are **activated** at the site of inflammation.
- **Most of the mediators are short-lived.**
- **One mediator can stimulate the release of other mediators.**

MEDIATORS OF INFLAMMATION

I. Cell-derived mediators:

- rapidly released from **intracellular granules** (e.g., amines) **or**
- Synthesized de novo (e.g., PGs, LKs, cytokines) in response to a stimulus.

II. Plasma-derived mediators:

- **E.g: Complement proteins.**
 - are present in the circulation as inactive precursors.
 - must be activated, usually by a series of proteolytic cleavages.
 - They are produced mainly in the liver.
 - are effective against circulating microbes.
 - can be recruited into tissues.

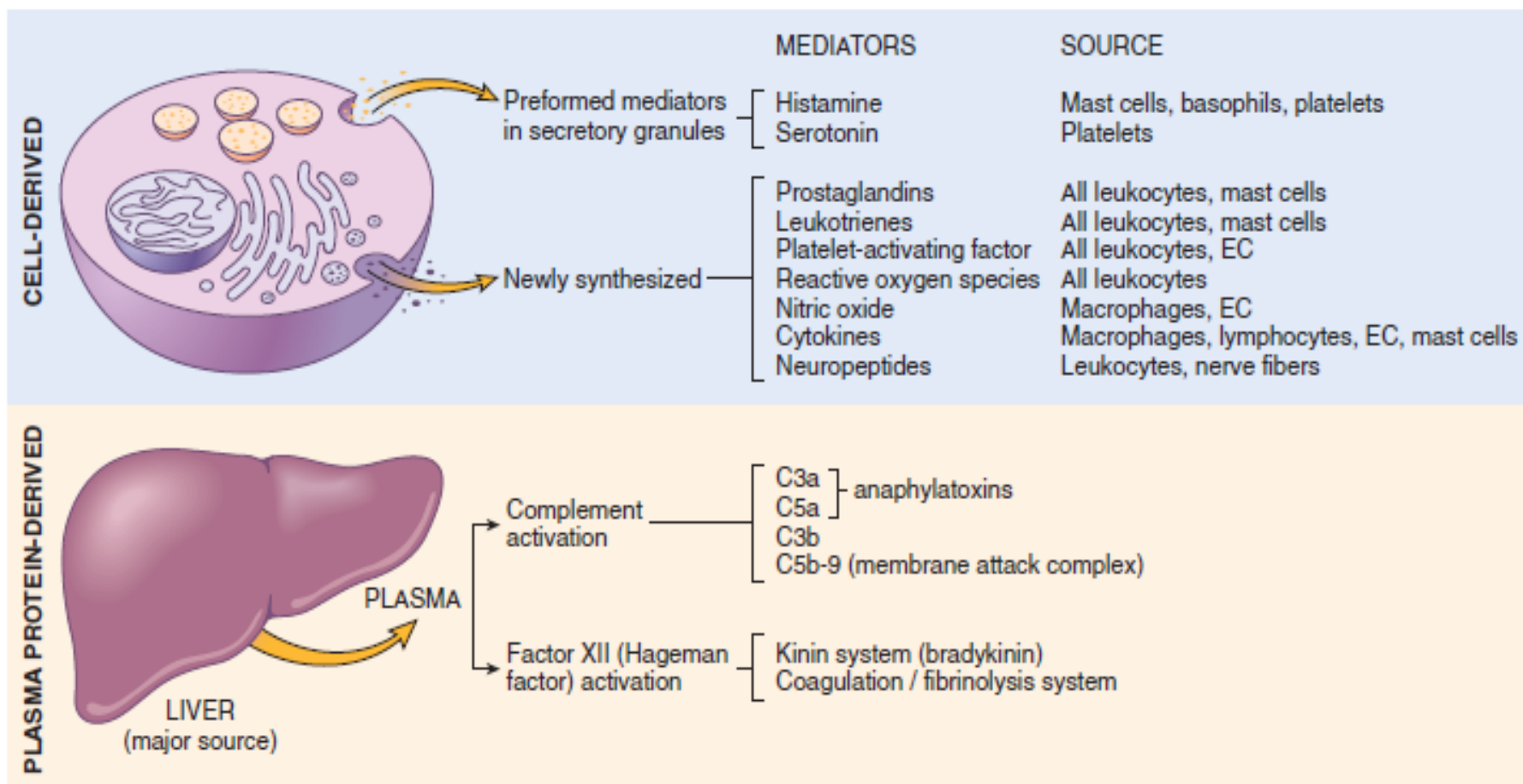


Figure 2-15 Mediators of inflammation. The principal cell-derived and plasma protein mediators are shown. EC, endothelial cells.

Table 3.8 Role of Mediators in Different Reactions of Inflammation

Reaction of Inflammation	Principal Mediators
Vasodilation	Histamine Prostaglandins
Increased vascular permeability	Histamine C3a and C5a (by liberating vasoactive amines from mast cells, other cells) Leukotrienes C ₄ , D ₄ , E ₄
Chemotaxis, leukocyte recruitment and activation	TNF, IL-1 Chemokines C3a, C5a Leukotriene B ₄
Fever	IL-1, TNF Prostaglandins
Pain	Prostaglandins Bradykinin
Tissue damage	Lysosomal enzymes of leukocytes Reactive oxygen species

I. Cell-Derived Mediators

I. Vasoactive Amines:

- They are stored as **preformed** molecules in cells.
- Therefore among the **first mediators** to be **released** during inflammation.

a. Histamine

- The **richest source** of this amine is the **mast cells**.
 - It is also found in **basophils** and **platelets**.
- Histamine causes:
 1. **Dilation** of the **arterioles**.
 2. Increases the **permeability** of venules.

b. Serotonin (5-hydroxytryptamine):

- is present in **platelets** (and enterochromaffin cells).
- It has actions similar to those of histamine.

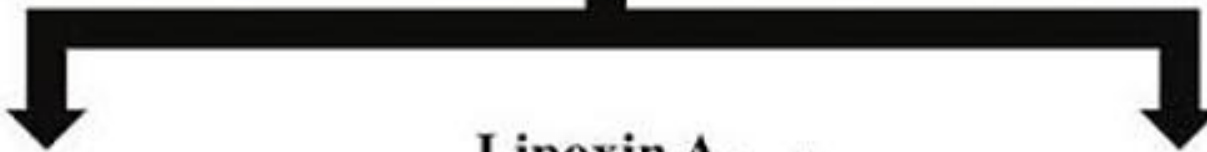
II. Arachidonic Acid Metabolites: (Prostaglandins, Leukotrienes and Lipoxins)

- Arachidonic acid (AA) is a fatty acid.
- Cell activation is accomplished by mechanical, chemical, and physical stimuli or other mediators (e.g., C5a).
- AA metabolites are synthesized by **two major classes of enzymes**:
 1. **Cyclooxygenases pathway (COX)**: produce prostaglandins (PGs) including thromboxane (TxA₂).
 2. **Lipoxygenases pathway**: that generate leukotrienes and lipoxins.

(Cell membrane)
PHOSPHOLIPIDS



ARACHIDONIC ACID



Prostaglandin G₂



Prostaglandin H₂



- Prostacyclin
- Thromboxane A₂
- Prostaglandin E₂

**Lipoxin A₄
Lipoxin B₄**



5-HPETE



Leukotrienes A₄ & B₄



- Leukotriene C₄
- Leukotriene D₄
- Leukotriene E₄

I. Prostaglandins

- Prostaglandins (PGs) are produced by **mast cells, macrophages, endothelial cells**, and many other cell types.
- They are generated by the actions of two cyclooxygenases called **COX-1** and **COX-2**.
- PGD2 with PGE2:
 1. causes vasodilation and increases the permeability of postcapillary venules.
 2. potentiating exudation and resultant edema.
- **PGD2** also is a **chemoattractant** for **neutrophils**.
- Systemic manifestation of inflammation:
 - prostaglandins are involved in the pathogenesis of **pain** and **fever**.
 - **PGE2** makes the skin hypersensitive to **painful stimuli**, and causes **fever** during infections.

II. Leukotrienes

- Leukotrienes are produced in leukocytes and mast cells, by the action of lipoxygenase.
- function:
 1. vascular and smooth muscle reactions and
 2. leukocyte recruitment.
- **LTB₄** is produced by neutrophils and some macrophages.
- **LTB₄** is a potent chemotactic agent and activator of neutrophils.
- Leukotriene **LTC₄ , LTD₄ and LTE₄**:
 1. are produced mainly in mast cells.
 2. cause intense vasoconstriction, bronchospasm (important in asthma), and increased permeability of venules.

III. Lipoxins

- Generated from arachidonic acid by the **lipoxygenase pathway**.
- The lipoxins suppress inflammation by:
 1. **Inhibiting** the recruitment of leukocytes.
 2. They inhibit neutrophil **chemotaxis**.
 3. Inhibit **neutrophil adhesion** to endothelium.

III. CYTOKINES AND CHEMOKINES

Cytokines:

- Are proteins produced by:
 1. activated lymphocytes, macrophages and dendritic cells.
 2. also endothelial, epithelial, and connective tissue cells.
- Function:
 - being involved in cellular immune responses.
 - play important roles in both acute and chronic inflammation.
- In the inflammatory response:
 - **Tumor Necrosis Factor (TNF)** and **Interleukin-1 (IL-1)**.
 - Their secretion can be stimulated by endotoxin, microbial products, immune complexes, and physical injury.
- The most important roles of these cytokines in inflammation are the following:
 - Endothelial activation.
 - Activation of leukocytes and other cells.
 - Systemic acute-phase response.
 - TNF regulates energy balance by promoting lipid and protein catabolism and by suppressing appetite.

Chemokines:

- Are a family of **small proteins**.
- Act primarily as **chemo-attractants** for specific types of leukocytes.
 - for e.g. **IL-8** acts primarily on **neutrophils**.
- Chemokines:
 - Bind to proteoglycans.
 - Displayed at high concentrations on the surface of endothelial cells and in the extracellular matrix.
- They have two main functions:
 1. Acute inflammation.
 2. Maintenance of tissue architecture.

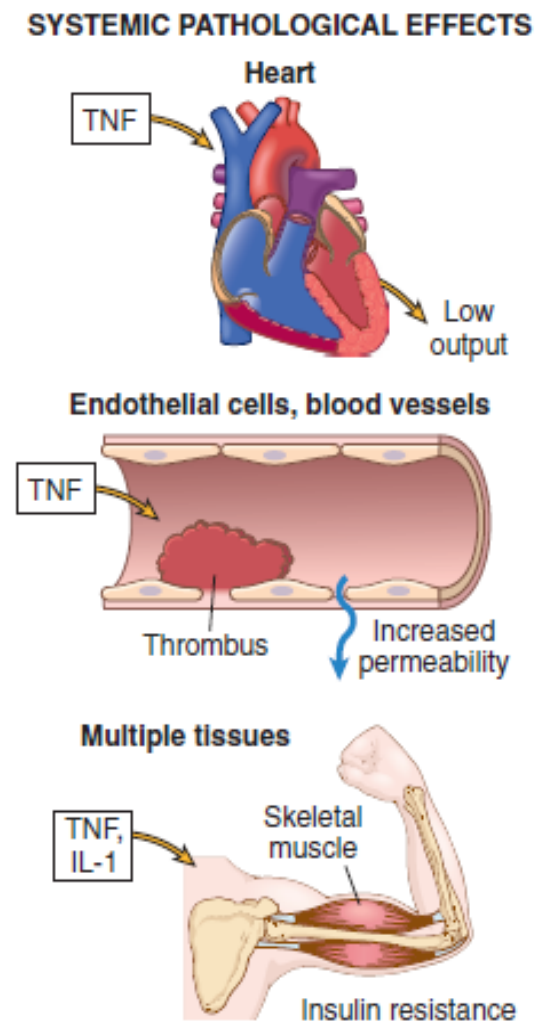
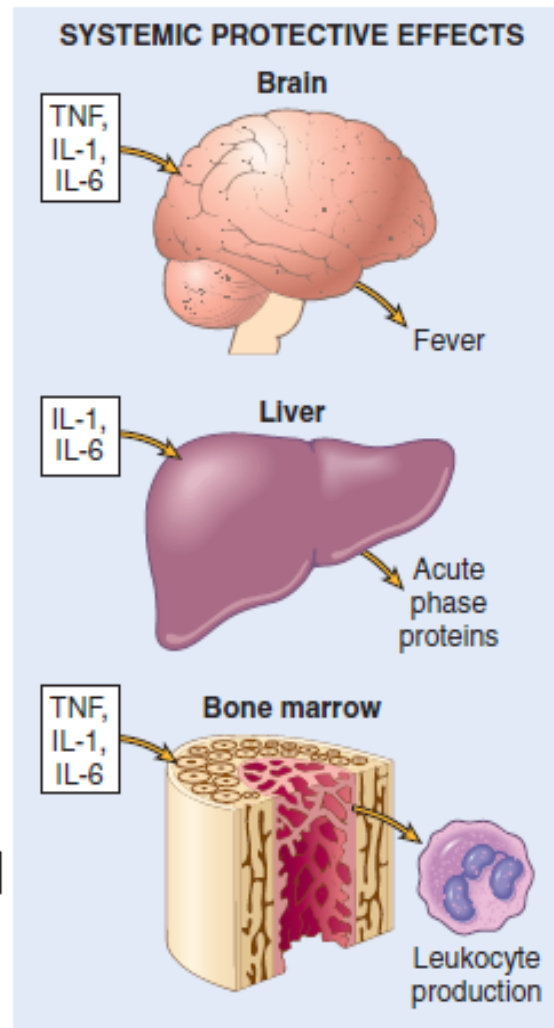
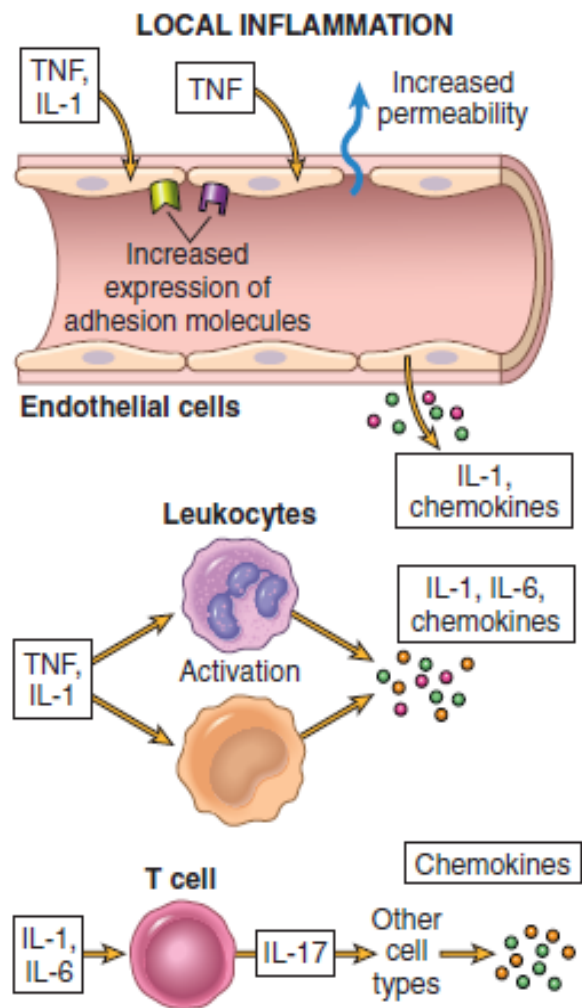


Fig. 3.10 Major roles of cytokines in acute inflammation. PDGF, Platelet-derived growth factor; PGE, prostaglandin E; PGI, prostaglandin I.

II. Plasma proteins

- These belong to three interrelated systems, the complement, kinin, and clotting systems.

A . The complement System

- The complement system is a collection of **soluble proteins** and their membrane **receptors**.
- Complement proteins are present in **inactive forms** in the **plasma**.
- many of Complement proteins are **activated** to become proteolytic enzymes that **degrade** other complement proteins, thus forming an enzymatic cascade capable of amplification.
- The **critical step** in complement activation is the **proteolysis** of the third (most abundant) component, **C3**.
- Three pathways of complement activation lead to the formation of an enzyme called the C3 convertase, which **splits C3** into two functionally fragments, **C3a** and **C3b**.

- In the process of complement activation, a number of *complement components* are elaborated to mediate a variety of phenomena in acute inflammation:

1. **Vascular phenomena:** C3a, C5a

- Histamine release from mast cells.
- Increase vascular permeability.
- Vasodilation .

2. **Chemo-attractants:** for e.g. C5a

1. **Opsonins:** C3

B . The kinin System

- Following the exposure of blood plasma to vascular basement membrane **collagen** after injury to endothelial cells.
- One of the important **kinins** is the **vasoactive bradykinin**.
 1. Bradykinin increases vascular permeability.
 2. contraction of smooth muscle.
 3. dilation of blood vessels.
 4. pain when injected into the skin.
- These effects are similar to those of histamine.

C . Clotting System

- **Activation** of the **clotting system** results in the formation of **thrombin**.
- Thrombin generates **insoluble fibrin clot**.
- Thrombin binds to specific receptors expressed on platelets, endothelial and smooth muscle cells, triggering recruitment of leukocytes

Thank

you

